

INTRODUCTION TO CLOUD COMPUTING



MOHANA MURALI GURUNATHAN
Work Integrated Learning Programmes

AGENDA

Cloud Deployment Models

Cloud Infrastructure and Deployment

Benefits, limitations, and adoption drivers



Work Integrated Learning Programmes

CLOUD DEPLOYMENT MODELS

Private Cloud

Public Cloud

Community Cloud

Hybrid Cloud.



It all boils down to the priorities of an organization, the funding available, the agility needed to complete, the benefits of the collaborative exercises etc.

CLOUD DEPLOYMENT MODELS

Private cloud - Cloud computing infrastructure built exclusively for a single enterprise. It is the next step in the evolution of a corporate data center of today where the infrastructure is shared within the enterprise.

Community Cloud - Cloud computing infrastructure shared by a community of multiple organizations that generally, have a common purpose. An example of a community cloud is OpenCirrus, which is a cloud computing research testbed intended to be used by universities and research institutions.

Public cloud - Cloud computing infrastructure owned by a cloud service provider that provides cloud services to the public for commercial purposes.

Hybrid clouds – Cloud computing infrastructure which can be considered as a mixture of the above deployments. For example, an enterprise may rent storage in a public cloud for handling peak demand. The combination of the enterprise's private cloud and the rented storage then is a hybrid cloud

Used for a single organization;
can be internally or externally
hosted

PRIVATE

Shared by several
organizations;
typically externally
hosted, but may be
can be internally
hosted by one of
the organizations

COMMUNITY

HYBRID

PUBLIC

Composition of two or more
clouds (private, community or
public) that remain unique
entities but are bound together,
offering the benefits of multiple
deployment models; is internally
& externally hosted

Provisioned for open
use for the public by a
particular organization
who also hosts the
service

PUBLIC CLOUD

High elasticity

Subscription-based

Low cost

Highly Scalable

No CapEx

The cloud vendor is the entity that is completely responsible for the public cloud. Develop, Manage, and Maintain the resource pool shared between multiple tenants from across the network.

Minimal Technical Control, Lack of Cost Control, Lack of Security

PRIVATE CLOUD

Customized Security

Efficient performance

High CapEx

Data center - on-premise / operated by a third-party vendor off-site. Computing resources are isolated and delivered via a secure private network. Nothing is shared with any other organization or entities.

- Highly regulated industries and government agencies
- Sensitive data
- High Control/ High Security wrt the infra and the workloads
- Large enterprises that require advanced data center technologies to operate efficiently and cost-effectively
- Organizations that can afford to invest in high performance and availability technologies (last but not the least)

High TCO, Scalability may become an issue for unpredictable workloads

PRIVATE CLOUD

What are the various possible flavours of this model?

What are the prominent examples of this model?

Who are the Key Players?

PRIVATE CLOUD - FLAVOURS

On-Prem Private Cloud

Hosted: Internally within the organization's own data centre

Managed: Internal IT team of the Organization.

Control: Complete control over hardware, security, and configuration.

Cost: High. Upfront capital expenditure (CapEx) for the hardware for most of the cases. Maintenance costs.

Managed Private Cloud

Hosted: On-premises or in a vendor/partner's data centre.

Managed: Entirely by the third-party service provider.

Control: High control, but operational management is outsourced.

Cost: Both possibilities exist – CapEx and OpEx – when it is On-prem. It is mostly OpEx when it is in Vendor/Partner's data centre

PRIVATE CLOUD - FLAVOURS

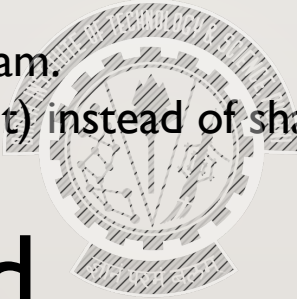
Hosted Private Cloud

Hosted: Third-party cloud provider's data centre.

Managed: Provider or the customer's internal team.

Control: Dedicated physical servers (single-tenant) instead of shared hardware.

Cost: Vast majority of deployments (only OpEx).



Virtual Private Cloud

Hosted: Within a public cloud infrastructure (e.g., AWS, Azure, Google Cloud).

Managed: By the public cloud provider.

Control: Isolated, private network environment carved out using virtual tools.

Cost: Highly scalable, pay-as-you-go model (OpEx)

PRIVATE CLOUD - FLAVOURS

The differences between Managed Vs Hosted Private Cloud – Do they look blurred?



Work Integrated Learning Programmes

PRIVATE CLOUD - FLAVOURS

The differences between Managed Vs Hosted Private Cloud – Do they look blurred?

Hosted Private Cloud - defined by **Location**. The infrastructure lives off-premises in a third-party data centre and not on the Premises of the organization.

Managed Private Cloud - defined by **Operations**. A third party handles the day-to-day maintenance (patching, monitoring, troubleshooting), regardless of where the hardware sits.

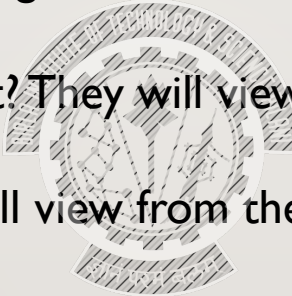
The blur: **Most Hosted Private Clouds are also Managed**. Providers rarely rent the space and hardware in their data centre without also offering to manage it for the organization (company) that has requested the space and hardware. When a hosted private cloud is also managed, then in business jargon/terminology the terms are used interchangeably. If the correct technical word/precision is desired, then the correct word would be “**Managed Hosted** Private Cloud”

PRIVATE CLOUD - FLAVOURS

When a hosted private cloud is also managed what will the

Sales/Procurement Team call it? They will view from the perspective of location/budget!

IT Operations call it? They will view from the perspective of Managing the operations!



Focus	Terminology Used	Reason
Location & Budget	Hosted Private Cloud	No cost for the facilities
Daily Tasks (Managing the operations)	Managed Private Cloud	No cost for the operations.

PROMINENT EXAMPLES OF PRIVATE CLOUD

Prominent examples of private cloud vary based on how the cloud is deployed

Is it built using dedicated software on-premises
or rented as isolated infrastructure from
external vendors?

PROMINENT EXAMPLES OF PRIVATE CLOUD

Open-Source & Enterprise Software (On-Prem)

These software platforms allow an organization's internal IT team to turn raw, bare-metal hardware inside their own data centre into a self-service private cloud.

OpenStack: The most famous open-source platform. It is heavily used by large enterprises and telecommunication companies to avoid vendor lock-in.

VMware Cloud Foundation: The industry standard for enterprise virtualization. It bundles vSphere, vSAN, and NSX to create highly secure corporate clouds.

Red Hat OpenShift: A Kubernetes-focused platform engineered for modern, containerized applications that need strict zero-trust security control.

PROMINENT EXAMPLES OF PRIVATE CLOUD

Hardware-Based & Managed Private Clouds

These are pre-configured physical infrastructure stacks or appliances delivered directly to your data center, often fully managed by the vendor.

HPE GreenLake: A consumption-based IT model where Hewlett Packard Enterprise sets up private cloud hardware in the customer's facility. Customer pays only for what they use.

Dell APEX Flex on Demand: Customized on-premises server and storage setups designed, deployed, and supported entirely by Dell.

Oracle Cloud@Customer: A private cloud system where Oracle places its database-optimized infrastructure inside the data centre of the customer for total data sovereignty.

PROMINENT EXAMPLES OF PRIVATE CLOUD

Hyperscaler Extensions (Hybrid/Hosted Private)

Major public cloud providers offer extensions that bring their exact public cloud software, APIs, and hardware into the physical facility of the customer.

AWS Outposts: Physical racks of Amazon Web Services servers installed directly in customer's local data centre. This is operated and fully managed by AWS.

Azure Stack Hub: Microsoft's ecosystem extension allowing a company to run cloud-native Azure apps and services entirely on-premises, completely disconnected from the internet if needed.

Google Distributed Cloud (Formerly Anthos): Google's hardware and software offering designed to run containerized services locally with integrated AI/ML tools.

PRIVATE CLOUD – KEY PLAYERS

Enterprise Software & Virtualization Leaders

These companies provide the foundational software architecture used to abstract bare-metal physical servers into secure, automated private cloud environments

Broadcom (VMware): The industry standard for traditional corporate data centres via **VMware Cloud Foundation**.

Nutanix: A top market leader specializing in hyperconverged infrastructure (HCI) software that seamlessly merges compute, storage, and networking.

IBM (Red Hat): Commands the enterprise container market using **Red Hat OpenShift** and **OpenStack** platforms for open-source private cloud management

PRIVATE CLOUD – KEY PLAYERS

Hardware & Managed Infrastructure Giants

Dell Technologies: Offers Dell APEX infrastructure, combining PowerEdge servers and PowerFlex software-defined storage for turnkey private and hybrid environments.

Hewlett Packard Enterprise (HPE): Drives hybrid and private adoption via the HPE GreenLake edge-to-cloud platform, delivering on-premises cloud infrastructure as a service.

Cisco Systems: Integrates data centre networking, compute, and unified management through Cisco Intersight for secure private clouds.

Lenovo Group: Supplies scalable ThinkAgileSystems hardware optimized for hyperconverged infrastructure and enterprise private cloud deployments.

HYBRID CLOUD

Additional Computing Capacity does not require high CapEx

Decide to be Homogenous or Heterogeneous with the Cloud

Cost Control

Policy Driven

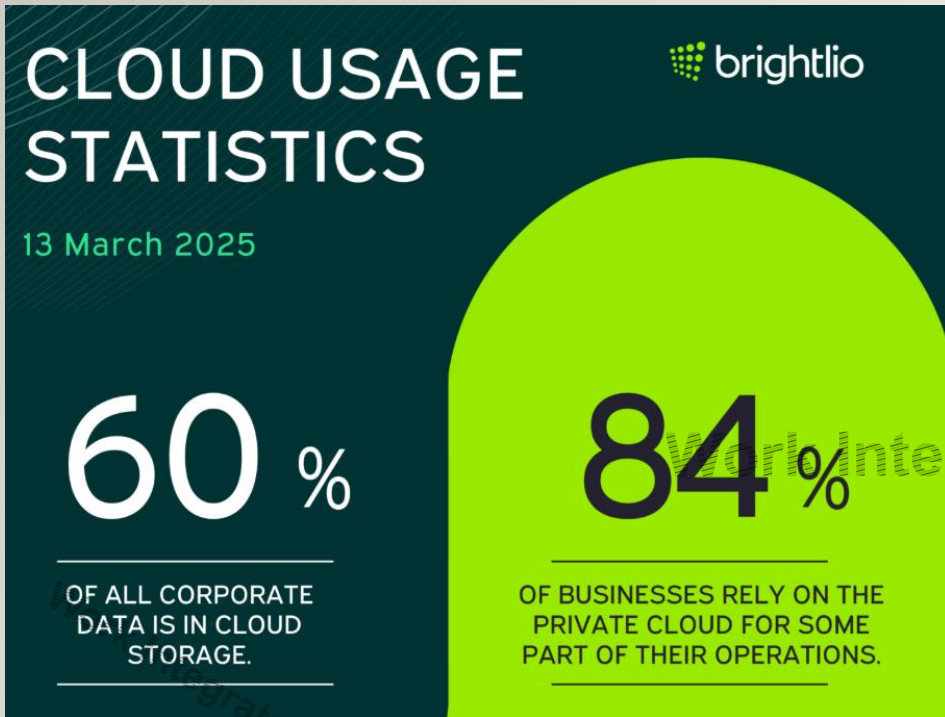
Scale with Security

Increase in reliability

- Organizations serving multiple verticals facing different IT security, regulatory, and performance requirements
- Optimizing cloud investments without any compromises
- Strategic approach to cloud investments - continuously switch and tradeoff between the best cloud service suited at the point in time.

Price tracking, Management, Added Complexity as the organization operates and manages a mix of resources

SOME OF THE INTERESTING FACTS FROM THE NET



50% of all data worldwide will be stored in the cloud by this year.
96% of companies use at least one public cloud
Over **60%** of all corporate data is in cloud storage.
The top priorities for clients used cloud based services are cloud spending (**82%**), security concerns (**79%**), lack of expertise (**78%**), and cloud migrations (**71%**).
85% of organizations will be cloud-first by 2025.
Just **2%** of customers use a single private cloud solution.

TOP CLOUD COMPUTING STATISTICS

With global spending on **cloud computing** predicted to reach **\$679 billion in 2024**, let's look at some key statistics around this area that may inform your business decisions.

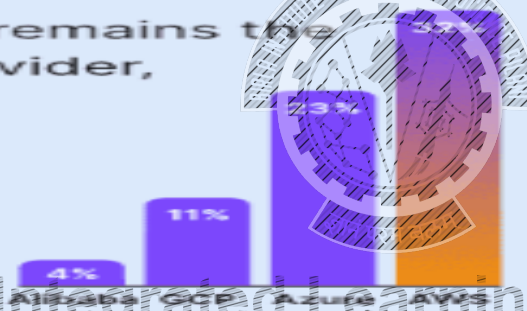
By 2025, there will be **200 zettabytes** (a trillion gigabytes) of **data** in the world.

By 2026, the cloud computing market is forecast to be worth **\$947.3 billion**.



96% of companies use **the public cloud**.

Amazon Web Services remains the biggest public cloud provider, with **32%** of the market.



84% of companies use **the private cloud**.

The main challenge facing cloud decision-makers is **managing cloud spend** (82%).



94% of businesses noted **improvements in their security** after moving to the cloud.



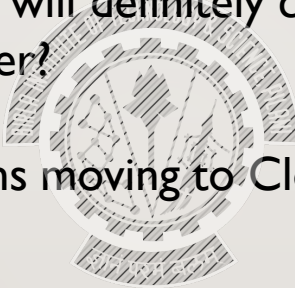
Reference: <https://spacelift.io/blog/cloud-computing-statistics>

BPS WLP | Generated: 09-Aug-2024 16:05:47

BENEFITS/LIMITATIONS/FACTS DRIVING THE ADOPTION

When an organization moves to the cloud, they will definitely consider the benefits and limitations.
What are the major aspects they should consider?

What are the major drivers for the organizations moving to Cloud?



Work Integrated Learning Programmes

CLOUD - BENEFITS

Cost Efficiency: Shifts capital expenditure (CapEx) to predictable (variable) operating expenses (OpEx).

Elastic Scalability: Allocates or shrinks compute power instantly based on real-time application demand.

High Availability(HA): Automated multi-region replication to guarantee robust disaster recovery.

Automatic Maintenance: Offloads system patching, infrastructure upgrades, and hardware refreshes to the vendor

Provisioning/Deploying with a Global Footprint: Provision Infra and deploy applications across multiple geographic regions within a few clicks.

On Demand, DR (Disaster Recovery)...

CLOUD - LIMITATIONS

Internet: Demands persistent, high-speed connectivity; outages entirely halt access to workloads. This is a **Dependency**

Vendor Lock-In: Introduces massive technical barriers when migrating data out due to proprietary ecosystem tools.

Hidden Costs: Accumulates unexpected fees through outbound data **egress** charges. Unnoticed / Unmanaged idle resources still needs to be paid!

Shared Responsibility: Divides data security duties, requiring complex governance to avoid customer-side misconfigurations.

Control Restrictions: Limits deep system customization because the underlying hardware architecture belongs to the Cloud provider.

CLOUD – INGRESS & EGRESS – DERIVED FROM LATIN

Ingress (entering) : In cloud computing, **data ingress** means data is "stepping into" the cloud network from the outside

egress (exiting) – In cloud computing, **data egress** means data is "stepping out" of the cloud network to an outside destination. Users may consider this as upload/download. But for the Cloud, data is either entering or exiting.

Ingress: Usually free. Providers want to make it as easy and cheap as possible for the data to be uploaded and stored

Egress: Billed per gigabyte. Moving data out uses external bandwidth and creates a financial barrier to switching providers or moving data back on-premises.

Average Egress rate for AWS: Roughly \$0.09 per GB for the first 10 TB (

Differs between cloud providers. It even differs between different regions for the same cloud provider.

Moving data within the same cloud provider (for example, replicating an AWS S3 bucket in Mumbai to Hyderabad) will cost less as it is inter-region data transfer. For example, for AWS this can be \$0.02 per GB rather than the average of \$0.09 per GB.

Moving data within the same cloud provider (but different continents) will definitely cost more than the inter-region data transfer

CLOUD – ADOPTION DRIVERS

Agility & Innovation: Grants immediate access to advanced technologies - AI, ML, and big data analytical tools.

Time-to-Market: Accelerates software development cycles by eliminating long hardware procurement delays.

Remote Workplace Support: Connects globally distributed workforces securely via universal data accessibility.

Sustainability: Reduces corporate carbon footprints by utilizing multi-tenant, energy-efficient cloud data centres.

Regulatory Compliance: Offers built-in frameworks satisfying regional data mandates like GDPR and HIPAA.

CLOUD – FURTHER READING/ INDEPENDENT STUDY

GDPR – General Data Protection Regulation – EU law to protect all personal data and privacy for people living in the EU or UK. Users can access their data and also can ask that the data be deleted (the “right to be forgotten”)

HIPAA – Health Insurance Portability & Accountability Act. Covers many aspects like who can view and share a person’s medical information. Includes protection and enforcement rules too.

Understand how cloud providers support GDPR, HIPAA etc. Would it be a shared responsibility or entirely on the cloud provider?

Understand how companies can benefit by moving to the cloud – from various aspects. For example, tax benefits. Does India provide any tax benefits. How are sustainable efforts like cloud computing enable companies to save in taxes/green house gas emissions etc? While the cloud does not completely eliminate the carbon footprint of IT, it is said that it drastically reduces it. Find out how it is possible and why it is possible etc?

Sky/Cloud is the limit for your research/independent study!